

Constructive Reduced Generators for Mean-Force Equilibrium

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We study when the reduced equilibrium state of an open quantum system admits a compact constructive generator rather than only an implicit density operator. Starting from the quenched representation of the traced Gibbs state, we derive an exact Gaussian reformulation in which bath physics enters through scalar kernel moments while operator growth is carried entirely by the adjoint chain generated by the bare Hamiltonian and the coupling. This yields a closure criterion: a finite closed-form Hamiltonian of mean force exists only when the adjoint-generated operator family closes inside the chosen ansatz. For the spin-boson qubit this criterion is satisfied exactly, and the reduced equilibrium compresses to a closed Bloch-plane generator controlled by two response channels and a single influence magnitude. The same variables organize the numerical crossover: once the dressed state outgrows the available basis, simulation error reflects model-class failure rather than a breakdown of the exact theory. The resulting perspective connects strong-coupling equilibrium physics to broader questions of reduced representation, interpretable model classes, and finite-resource inference.

INTRODUCTION

Recent work in the synthetics program has argued that representations should be treated as physical objects, with cost attached to the distinctions they preserve and the coarse-grainings they stabilize [1]. A related intuition appears in machine learning, where compressed internal variables are valuable only insofar as they retain task-relevant structure [2]. Open-system equilibrium provides a particularly sharp physics instance of the same problem: after reducing a large interacting system to a smaller subsystem, when does the result still admit a compact generator rather than only a black-box density matrix?

In closed quantum statistical mechanics, equilibrium is generated by a Hamiltonian, $\rho \propto e^{-\beta H}$. For an open system with finite coupling, the operational equilibrium state of the subsystem is the reduced state of the global Gibbs ensemble,

$$\begin{aligned} \bar{\rho}_Q(\beta) &= \text{Tr}_X e^{-\beta H_{\text{tot}}}, \\ e^{-\beta H_{\text{MF}}(\beta)} &\propto \frac{\text{Tr}_X e^{-\beta H_{\text{tot}}}}{Z_X(\beta)}, \quad Z_X(\beta) = \text{Tr}_X e^{-\beta H_X}. \end{aligned} \quad (1)$$

The Hamiltonian of mean force (HMF) is the operator that restores a Gibbs-form description after the bath has been traced out. It is therefore the exact reduced equilibrium generator whenever system–bath coupling is finite [3–8].

Why care about a mean-force Hamiltonian at all? First, it underlies the free-energy, work, and entropy relations of strong-coupling thermodynamics [3, 6, 9–12]. Second, it provides a consistent reduced equilibrium state when correlations with the environment cannot be ignored [5, 13, 14]. Third, and more conceptually, it turns a traced Gibbs operator into a constructive reduced description that can be inspected, truncated, or compared across operator families. In that limited sense the HMF problem is close in spirit to a mechanistic-interpretability

question: can the effect of a large hidden system be summarized by a small explicit internal generator rather than only by its output state?

The literature is broad but structurally clear. Canonical definitions and thermodynamic identities are now well established [3–6, 9]. Exact evaluations exist in special cases: commuting interactions close trivially, quadratic models preserve Gaussianity, and finite-dimensional examples such as the spin-boson model provide controlled testbeds [15–19]. Benchmark asymptotes in weak coupling and in strongly dressed regimes are also known and remain important checks on any exact construction [20, 21]. Recent work has further emphasized operator structure, bath feedback, and finite-reservoir generalizations [22–24].

Outside solvable models, however, the HMF is usually accessed numerically or perturbatively. Reaction-coordinate mappings, polaron transforms, hierarchical equations, stochastic approaches, and direct imaginary-time solvers can compute $\bar{\rho}_Q(\beta)$, but they rarely return a compact closed-form generator [25–36]. The obstruction is therefore not the existence of the reduced equilibrium state but its representability: when does the logarithm of a traced exponential stay inside a restricted operator family such as few-body terms, local operators, or a chosen algebra?

The influence-functional formalism is a natural language for this question. For Gaussian baths with linear coupling it provides an exact route to integrating out environmental degrees of freedom [16, 17, 37]. In equilibrium it becomes an imaginary-time influence functional, which admits a Hubbard–Stratonovich rewriting as a quenched stochastic average [38–40]. This formulation makes the reduction step explicit: the partial trace becomes an average over imaginary-time back-action histories, while all remaining questions are pushed onto the system operator algebra.

The present paper takes the quenched-density con-

struction of Ref. [41] as its exact starting point and uses it to answer a representability question. For Gaussian baths with linear coupling, bath statistics collapse to scalar kernel moments while operator growth is generated entirely by the adjoint chain $\{\text{ad}_{H_Q}^n(f)\}_{n \geq 0}$. This yields a closure criterion: a finite closed-form HMF exists only when the adjoint-generated operator family closes inside the target ansatz. We then instantiate this criterion in the spin-boson qubit, where the reduced equilibrium compresses to a closed Bloch-plane generator controlled by two response channels and a single influence magnitude. The same variables organize the numerical crossover, revealing a representability bottleneck once the dressed reduced state outgrows the chosen finite basis.

QUENCHED REPRESENTATION AND INFLUENCE FUNCTIONAL

We begin by recapitulating the quenched representation introduced in Ref. [41]. To proceed directly from Sec. , we explicit the composite model. We denote the bare system Hamiltonian by H_Q and write

$$H_{\text{tot}} = H_Q + H_X + H_{\text{int}}, \quad (3)$$

where H_X is the bath Hamiltonian. We assume a factorizable interaction

$$H_{\text{int}} = f \otimes B, \quad (4)$$

where f acts on the system and B is a bath operator. We can define the reduced equilibrium operator (up to normalisation) by $\bar{\rho}_Q(\beta) \equiv \text{Tr}_X e^{-\beta H_{\text{tot}}}$. As shown in Ref. [41], this can be represented as a *quenched density*. This is an average over a stochastic propagator, given by:

$$\bar{\rho}_Q(\beta) = \mathbb{E}_\xi [U_\xi(\beta)], \quad (5)$$

$$U_\xi(\beta) \equiv \mathcal{T}_\tau \exp \left[- \int_0^\beta d\tau (H_Q + \xi(\tau)f) \right], \quad (6)$$

where $\xi(\tau)$ is a stochastic process whose statistics encode the bath correlations, and $\mathcal{T}_\tau$ denotes time-ordering in imaginary time. Regardless of the precise form of bath and coupling, the reduced density must always be describable in this form.

This result can be understood intuitively by first observing that since the environment influence can only enter through the system coupling f , its influence can be captured by attaching a τ -dependent driving field to f [42, 43]. If this were a single deterministic field however, it would correspond to the bath exerting the *same* back-action history for every microscopic bath configuration. But Tr_X averages over many bath microstates in the thermal ensemble, and hence over many back-action histories. In this sense the partial trace is necessarily an average over histories in imaginary time, and

the quenched representation simply makes this averaging explicit. For each realisation $\xi(\tau)$ the system evolves under an imaginary-time Hamiltonian $H_Q + \xi(\tau)f$; the bath is then recovered by averaging over $\xi(\tau)$ with a law chosen to reproduce the bath-induced correlations. In this sense $\xi(\tau)$ is not a physical external control field but an efficient parametrisation of the bath history $B(\tau)$ as seen through the coupling channel.

We may understand what formal properties are demanded of ξ by considering the influence functional it is required to match. Working in imaginary time, introduce the bath interaction picture:

$$B(\tau) \equiv e^{\tau H_X} B e^{-\tau H_X}, \quad \tau \in [0, \beta], \quad (7)$$

and define the bath thermal state $\rho_X \equiv e^{-\beta H_X} / Z_X$. For an arbitrary c-number source $j(\tau)$ coupled linearly to $B(\tau)$, the bath generates a (time-ordered) functional

$$\mathcal{Z}_X[j] \equiv \text{Tr}_X \left[\mathcal{T}_\tau \exp \left(- \int_0^\beta d\tau j(\tau) B(\tau) \right) \rho_X \right]. \quad (8)$$

Here $j(\tau)$ is introduced as an external c-number source used to generate ordered bath correlators by functional differentiation. More generally, $j(\tau)$ may be any object commuting with the bath algebra (e.g. a system operator tensored with $\mathbb{I}_X$). In the influence-functional derivation it is ultimately supplied by the system history, which becomes a c-number function in the path-integral representation. In the influence-functional approach, tracing out the bath produces precisely such a functional, evaluated on the system history through the coupling channel.

From this, the bath contribution to the effective Euclidean action can be written as [16, 17, 37]

$$\mathcal{F}[f] \equiv \mathcal{Z}_X[f], \quad \Phi[f] \equiv \log \mathcal{F}[f]. \quad (9)$$

A key structural fact is that $\Phi[f] = \log \mathcal{F}[f]$ is the *cumulant generating functional* of the bath operator $B(\tau)$ with respect to the thermal state. Concretely, the generalised (time-ordered) cumulant theorem implies the connected expansion [44, 45]

$$\begin{aligned} \Phi[f] &= \sum_{n \geq 1} \frac{(-1)^n}{n!} \int_0^\beta d\tau_1 \cdots d\tau_n \\ &\quad \times K^{(n)}(\tau_1, \dots, \tau_n) f(\tau_1) \cdots f(\tau_n), \end{aligned} \quad (10)$$

where the kernels $K^{(n)}$ are the *connected* (cumulant) bath correlators, defined via [46]

$$K^{(n)}(\tau_1, \dots, \tau_n) \equiv \langle \mathcal{T}_\tau B(\tau_1) \cdots B(\tau_n) \rangle_c. \quad (11)$$

The explicit connection back to Eq. (8) is given via its functional differentiation:

$$\langle \mathcal{T}_\tau B(\tau_1) \cdots B(\tau_n) \rangle_c = (-1)^n \frac{\delta^n \log \mathcal{Z}_X[j]}{\delta j(\tau_1) \cdots \delta j(\tau_n)} \Big|_{j=0}. \quad (12)$$

The bath influence is then completely characterised by the hierarchy $\{K^{(n)}\}_{n \geq 1}$.

To connect the influence functional back to the quenched density, we use the fact that the bath influence depends on the system history only through the linear functional $\int_0^\beta d\tau f(\tau)B(\tau)$. One may therefore represent $\mathcal{F}[f]$ as the (generalised) characteristic functional of an auxiliary field $\xi(\tau)$ [38–40]:

$$\mathcal{F}[f] = \mathbb{E}_\xi \left[\exp \left(- \int_0^\beta d\tau \xi(\tau) f(\tau) \right) \right], \quad (13)$$

where $\mathbb{E}_\xi[\cdot]$ denotes averaging with respect to a (possibly complex) measure on ξ -histories chosen such that Eq. (13) holds.

Since $\xi(\tau)$ is a commuting c -number field, its n -point moments are symmetric under permutations of the time arguments. The influence kernels $K^{(n)}$ fix the *cumulants* of ξ via

$$\langle \xi(\tau_1) \cdots \xi(\tau_n) \rangle_c = K^{(n)}(\tau_1, \dots, \tau_n). \quad (14)$$

Consequently, the ordinary correlation functions of the noise are obtained from $\{K^{(m)}\}$ by the standard moment-cumulant relations:

$$\langle \xi(\tau_1) \cdots \xi(\tau_n) \rangle = \sum_{\pi \in \mathcal{P}_n} \prod_{C \in \pi} K^{(|C|)}(\{\tau_i\}_{i \in C}), \quad (15)$$

where $\mathcal{P}_n$ denotes the set of all partitions of the index set $\{1, \dots, n\}$, and C are the disjoint blocks of a given partition $\pi \in \mathcal{P}_n$. After shifting the mean so that $K^{(1)}(\tau) = \langle \xi(\tau) \rangle = 0$, one has (for example)

$$\langle \xi(\tau_1) \xi(\tau_2) \rangle = K^{(2)}(\tau_1, \tau_2), \quad (16)$$

$$\langle \xi(\tau_1) \xi(\tau_2) \xi(\tau_3) \rangle = K^{(3)}(\tau_1, \tau_2, \tau_3), \quad (17)$$

$$\begin{aligned} \langle \xi(\tau_1) \xi(\tau_2) \xi(\tau_3) \xi(\tau_4) \rangle &= K^{(4)}(\tau_1, \tau_2, \tau_3, \tau_4) \\ &\quad + K^{(2)}(\tau_1, \tau_2) K^{(2)}(\tau_3, \tau_4) \\ &\quad + K^{(2)}(\tau_1, \tau_3) K^{(2)}(\tau_2, \tau_4) \\ &\quad + K^{(2)}(\tau_1, \tau_4) K^{(2)}(\tau_2, \tau_3). \end{aligned} \quad (18)$$

QUENCHED DENSITY AND THE HAMILTONIAN OF MEAN FORCE

The Hamiltonian of mean force is, by definition, the operator whose Gibbs form reproduces the system's reduced equilibrium state. Equivalently, it is the *operator logarithm* of the unnormalised reduced equilibrium operator

$$\bar{\rho}_Q(\beta) \equiv \text{Tr}_X e^{-\beta H_{\text{tot}}}, \quad H_{\text{MF}}(\beta) \equiv -\frac{1}{\beta} \log \bar{\rho}_Q(\beta), \quad (19)$$

defined up to an additive multiple of the identity (fixed only when normalising the state).

The quenched representation in Eq. (5) supplies an exact stochastic parametrisation of the unnormalised mean-force Gibbs operator $\bar{\rho}_Q(\beta)$ by making the bath trace an explicit average over imaginary-time back-action histories. Combining Eq. (5) with Eq. (19) yields

$$H_{\text{MF}}(\beta) = -\frac{1}{\beta} \log \mathbb{E}_\xi [U_\xi(\beta)]. \quad (20)$$

Thus, constructing the mean-force Hamiltonian reduces to evaluating a stochastic average and then compressing the result via an operator logarithm. The conditions for being able to perform this average exactly is the focus of the present work.

To make the handling of this problem more concrete, we shall specialise the environment to the Caldeira-Leggett model, where the bath is a collection of harmonic oscillators ($H_X = \sum_k \omega_k b_k^\dagger b_k$) and the coupling is linear in bath coordinates ($B = \sum_k c_k x_k$). None of the results that follow are essentially dependent on this choice, and a generalisation to anharmonic environments is (relatively) straightforward. In the interests of comprehensibility however, we restrict our scope to quadratic environments. In this setting, the auxiliary field $\xi(\tau)$ becomes a stationary zero-mean Gaussian process completely characterized by its covariance

$$\mathbb{E}_\xi [\xi(\tau) \xi(\tau')] = K(\tau - \tau'). \quad (21)$$

The kernel $K(\tau)$ is determined by the bath spectral density $J(\omega) = \frac{\pi}{2} \sum_k \frac{c_k^2}{m_k \omega_k} \delta(\omega - \omega_k)$ via the relation [47]

$$K(\tau) = \frac{1}{\pi} \int_0^\infty d\omega J(\omega) \frac{\cosh(\omega(\beta/2 - |\tau|))}{\sinh(\beta\omega/2)}. \quad (22)$$

A key property of this kernel is its integrated strength. Integrating Eq. (22) yields

$$\begin{aligned} \int_0^\beta d\tau K(\tau) &= \frac{1}{\pi} \int_0^\infty d\omega J(\omega) \int_0^\beta d\tau \frac{\cosh(\omega(\beta/2 - |\tau|))}{\sinh(\beta\omega/2)} \\ &= \frac{1}{\pi} \int_0^\infty d\omega J(\omega) \frac{2}{\omega} \\ &= 2\lambda, \end{aligned} \quad (23)$$

where λ is the explicit reorganisation energy. Consequently, the total variance of the integrated noise field $\Xi = \int_0^\beta d\tau \xi(\tau)$ grows linearly with inverse temperature:

$$\mathbb{E}_\xi [\Xi^2] = \int_0^\beta d\tau \int_0^\beta d\tau' K(\tau - \tau') = 2\beta\lambda. \quad (24)$$

In the case that the system and its coupling commute, $[H_Q, f] = 0$ and f is τ -independent in imaginary time. In this instance time ordering drops out, and the average is given by [41]:

$$\bar{\rho}_Q(\beta) = \exp \left[-\beta \left(H_Q - \frac{\kappa_0(\beta)}{2} f^2 \right) \right], \quad (25)$$

which in turn yields

$$H_{\text{MF}}(\beta) = H_Q - \frac{\kappa_0(\beta)}{2} f^2 + \frac{1}{\beta} \log Z_X(\beta) \mathbb{I}. \quad (26)$$

Notably this correction to $H_{\text{MF}}(\beta)$ is entirely *classical* [41]. This is hardly surprising, but emphasises that quantum effects *must* stem from non-commutativity. A truly quantum theory of thermodynamics therefore only makes meaningfully distinct predictions when $[H_Q, f] \neq 0$. In this case however, the noise enters through a noncommuting operator inside $\mathcal{T}_\tau$, rendering the question of averaging highly non-trivial. In the next section, we attack this problem directly, deriving conditions under which $H_{\text{MF}}(\beta)$ possesses a closed form.

GENERATING-FUNCTION STRUCTURE OF THE INFLUENCE OPERATOR

Quenched propagator and Gaussian resummation

The starting point is the quenched propagator introduced in Sec. . In the imaginary-time interaction picture with respect to H_Q , the unnormalised reduced equilibrium operator takes the form

$$\bar{\rho}_Q(\beta) = e^{-\beta H_Q} \langle W_\xi(\beta) \rangle_\xi, \quad (27)$$

$$W_\xi(\beta) \equiv \mathcal{T}_\tau \exp \left[- \int_0^\beta d\tau \xi(\tau) \tilde{f}(\tau) \right],$$

where $\tilde{f}(\tau) = e^{\tau H_Q} f e^{-\tau H_Q}$ and the noise covariance is $\langle \xi(\tau) \xi(\tau') \rangle_\xi = K(\tau - \tau')$.

Because the bath is Gaussian, only the second cumulant of ξ is non-vanishing. Time-ordering introduces a non-trivial operator structure, but the key identity—proved in Appendix A—is that the Wick resummation of all pairings exponentiates exactly:

$$\langle W_\xi(\beta) \rangle_\xi = \exp \left[\frac{1}{2} \int_0^\beta d\tau \int_0^\beta d\tau' K(\tau - \tau') \mathcal{T}_\tau [\tilde{f}(\tau) \tilde{f}(\tau')] \right]. \quad (28)$$

The proof is a direct application of Wick's theorem to the Dyson expansion of W_ξ : at order $2p$ in the expansion, there are $(2p)!/(2^p p!)$ distinct Wick pairings, each weighted by the two-point kernel $K(\tau_i - \tau_j)$. The prefactors from the Dyson expansion precisely cancel the pairing multiplicity, leaving only the singly-connected pairs.

We can exploit the symmetry $K(\tau - \tau') = K(\tau' - \tau)$ to fold the square domain onto the triangular one (see Appendix A), such that the exponent simplifies to

$$\Delta(\beta) \equiv \int_0^\beta d\tau \int_0^\tau d\tau' K(\tau - \tau') \tilde{f}(\tau) \tilde{f}(\tau'), \quad (29)$$

and the averaged reduced state takes the product form

$$\bar{\rho}_Q(\beta) = e^{-\beta H_Q} e^{\Delta(\beta)}. \quad (30)$$

The influence exponent $\Delta(\beta)$ is quadratic in $\tilde{f}$ and depends on the bath exclusively through the two-point kernel K . Here the absence of higher-order bath objects is not an approximation, but an exact consequence of the Gaussian nature of the environment.

The ordered Green function and frequency compression

At this stage, the operator Δ is still formally defined as an integral over the bath degrees of freedom. There is however a particularly satisfying interpretation of this operator in which the effect of the bath is readily understood. To make this explicit, we work in the eigenbasis of H_Q . Let $H_Q|i\rangle = E_i|i\rangle$, so that the matrix elements of the interaction-picture operator are

$$\tilde{f}_{ij}(\tau) = \langle i|\tilde{f}(\tau)|j\rangle = f_{ij} e^{\omega_{ij}\tau}, \quad \omega_{ij} \equiv E_i - E_j. \quad (31)$$

Substituting into Eq. (29) and expanding in the eigenbasis, the influence exponent takes the spectral form

$$\Delta(\beta) = \sum_{i,\ell} \Delta_{i\ell} |i\rangle\langle\ell|, \quad (32)$$

where evaluating the matrix elements of the operator product yields

$$\Delta_{i\ell} = \sum_j f_{ij} f_{j\ell} G^>(\omega_{ij}, \omega_{j\ell}), \quad (33)$$

where all bath and temperature dependence has been absorbed into the *ordered Green function*

$$G^>(\omega_1, \omega_2) \equiv \int_0^\beta d\tau \int_0^\tau d\tau' K(\tau - \tau') e^{\omega_1\tau + \omega_2\tau'}. \quad (34)$$

In standard finite-temperature perturbation theory, analogous time-ordered integrals are evaluated using discrete Matsubara frequencies. Here, the integration instead projects the bath response directly onto the Laplace frequencies set by the system's own energy differences; the superscript $>$ records that $\tau > \tau'$ throughout the integration domain.

The double integral in Eq. (34) collapses to a one-dimensional Laplace transform upon changing variables to $u = \tau - \tau'$ and $v = \tau'$:

$$G^>(\omega_1, \omega_2) = \frac{e^{\beta\omega_1} \mathcal{K}(\omega_2) - \mathcal{K}(\omega_1)}{\omega_1 + \omega_2}, \quad (35)$$

where

$$\mathcal{K}(\omega) \equiv \int_0^\beta du K(u) e^{\omega u} \quad (36)$$

is the bare Laplace transform of the bath kernel. Eq. (35) is an exact closed form valid for all $\omega_1 + \omega_2 \neq 0$. Importantly, it implies that for any transition between levels in

the reduced system, the entirety of the bath's influence is encoded by $\mathcal{K}$ evaluated at the transition frequency.

The resonant channel $\omega_1 + \omega_2 = 0$ requires separate treatment. Taking the limit carefully yields

$$R(\omega) \equiv G^>(\omega, -\omega) = \int_0^\beta (\beta - u) K(u) e^{\omega u} du, \quad (37)$$

where the factor $(\beta - u)$ is the available imaginary time after a bath correlation at separation u ; the resonant kernel thus accumulates bath memory over the full thermal interval, regulated by the finiteness of β .

The structure of Eq. (33) admits a pleasingly simple diagrammatic reading, illustrated in Fig. 1. To isolate the physically distinct effects of the bath, we partition the influence exponent into its secular (diagonal) and mixing (off-diagonal) components, $\Delta = \Delta_{\parallel} + \Delta_{\perp}$.

The secular sector $\Delta_{\parallel} \equiv \sum_i \Delta_{ii} |i\rangle\langle i|$ singles out the *closed loops* $i \rightarrow j \rightarrow i$, for which $\omega_{ij} + \omega_{ji} = 0$ identically. The resonance condition forces the bath propagator onto $R(\omega_{ij})$, the object that accumulates bath memory over the whole thermal interval. The diagonal elements are thus the exact, non-perturbative *self-energy* dressing of the i -th energy level by all levels j it is coupled to:

$$\Delta_{ii} = \sum_j |f_{ij}|^2 R(\omega_{ij}). \quad (38)$$

Conversely, the mixing sector $\Delta_{\perp} \equiv \sum_{i \neq \ell} \Delta_{i\ell} |i\rangle\langle \ell|$ describes *open chains* $i \rightarrow j \rightarrow \ell$, where energy is not conserved at the intermediate vertex. Using the closed form in Eq. (35), these components can be expressed directly in terms of the Laplace-transformed bath kernel $\mathcal{K}(\omega)$:

$$\Delta_{i\ell} = \frac{1}{\omega_{i\ell}} \sum_j f_{ij} f_{j\ell} [e^{\beta \omega_{ij}} \mathcal{K}(\omega_{j\ell}) - \mathcal{K}(\omega_{ij})] \quad (i \neq \ell), \quad (39)$$

which encodes the coherent transitions between distinct energy levels mediated by the asymmetric fluctuations of the bath.

KMS symmetry and the symmetrised reduced state

The bath kernel $K(\tau)$ satisfies the KMS periodicity condition $K(\tau) = K(\beta - \tau)$, which in Laplace space reads

$$\mathcal{K}(-\omega) = e^{-\beta \omega} \mathcal{K}(\omega). \quad (40)$$

Substituting into the closed form Eq. (35) and simplifying gives a corresponding symmetry of the ordered Green function:

$$G^>(-\omega_2, -\omega_1) = e^{-\beta(\omega_1 + \omega_2)} G^>(\omega_1, \omega_2). \quad (41)$$

Because transition frequencies telescope, $\omega_{ij} + \omega_{j\ell} = \omega_{i\ell}$, applying Eq. (41) to the matrix-element formula Eq. (33) immediately yields the *detailed-balance* constraint

$$\Delta_{i\ell} = e^{-\beta \omega_{i\ell}} \Delta_{\ell i}^*, \quad (42)$$

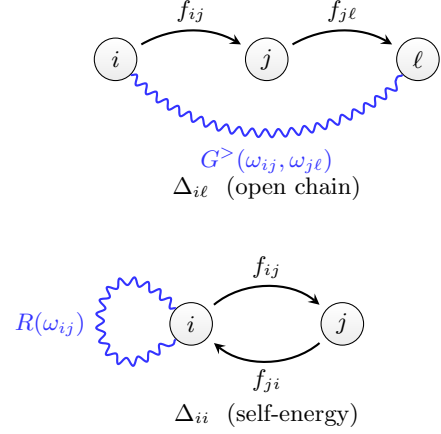


FIG. 1. Diagrammatic structure of the influence exponent $\Delta(\beta)$ in the H_Q eigenbasis. *Top*: The off-diagonal element $\Delta_{i\ell}$ (open chain) sums intermediate sites j , weighted by the bath propagator $G^>(\omega_{ij}, \omega_{j\ell})$ (wavy line) and two coupling insertions f (arrows). Energy is not conserved at the intermediate vertex. *Bottom*: The diagonal element Δ_{ii} (self-energy) is a closed loop $i \rightarrow j \rightarrow i$; the resonant condition $\omega_1 + \omega_2 = 0$ is satisfied, and the bath propagator collapses to $R(\omega_{ij})$.

or in operator form $\Delta^\dagger = \Pi \Delta \Pi^{-1}$ where $\Pi \equiv e^{-\beta H_Q}$. Although Δ itself is not self-adjoint, this relation is a precise constraint: the product $\Pi \Delta$ is Hermitian.

This symmetry allows us to recast the unnormalised reduced state Eq. (30) into a manifestly positive, Hermitian form. Writing the bare Gibbs factor as $\Pi \equiv e^{-\beta H_Q}$ gives $\bar{\rho}_Q = \Pi e^\Delta$. By splitting $\Pi = \Pi^{1/2} \Pi^{1/2}$ and inserting the identity $\Pi^{-1/2} \Pi^{1/2} = \mathbf{1}$, we symmetrically distribute the bare weight:

$$\bar{\rho}_Q = \Pi^{1/2} (\Pi^{1/2} e^\Delta \Pi^{-1/2}) \Pi^{1/2}. \quad (43)$$

Applying the operator identity $A e^B A^{-1} = \exp(ABA^{-1})$ to the inner bracket yields

$$\bar{\rho}_Q = \Pi^{1/2} e^S \Pi^{1/2}, \quad S \equiv \Pi^{1/2} \Delta \Pi^{-1/2}, \quad (44)$$

where S is the *symmetrised influence operator*. Its matrix elements evaluate to $S_{i\ell} = e^{-\beta \omega_{i\ell}/2} \Delta_{i\ell}$, which by Eq. (42) satisfy $S_{i\ell}^* = S_{\ell i}$. Thus, the modified exponent S is strictly self-adjoint. The partition function of the reduced system follows by cyclicity of the trace:

$$Z_Q \equiv \text{Tr}[\bar{\rho}_Q] = \text{Tr}[\Pi e^S] = \sum_i e^{-\beta E_i} (e^S)_{ii}. \quad (45)$$

Since S is self-adjoint, $(e^S)_{ii} \geq e^{S_{ii}} = e^{\Delta_{ii}}$ by Jensen's inequality applied to the spectral weights $|\langle i|n\rangle|^2$, giving the bound

$$Z_Q \geq \sum_i e^{-\beta E_i + \Delta_{ii}}, \quad (46)$$

where the right-hand side is a Boltzmann sum with bath-shifted energies and equality holds when S is diagonal

(i.e. when H_Q and Δ share an eigenbasis). The full result $Z_Q = \text{Tr}[\Pi e^S]$ captures all level-mixing corrections through the off-diagonal elements of S .

CLOSURE OF THE HAMILTONIAN OF MEAN FORCE

The obstruction to writing $H_{\text{MF}}(\beta)$ in a compact operator form is not the existence of the mean-force object (it is defined by a logarithm), but the *representability* of that logarithm inside a restricted operator family (few-body, local, Pauli strings, etc.). In the harmonic case this representability question reduces to a precise closure problem. To show this, we first write the quenched propagator in the imaginary-time interaction picture with respect to H_Q :

$$U_\xi(\beta) = e^{-\beta H_Q} W_\xi(\beta),$$

$$W_\xi(\beta) \equiv \mathcal{T}_\tau \exp \left[- \int_0^\beta d\tau \xi(\tau) \tilde{f}(\tau) \right], \quad (47)$$

where $\tilde{f}(\tau) \equiv e^{\tau H_Q} f e^{-\tau H_Q}$.

Because the noise is classical Gaussian with zero mean, the average of the time-ordered exponential resums exactly in terms of the second cumulant. Time ordering adds a non-trivial complication, which is addressed in Appendix A. The result may be succinctly expressed in terms of an *influence operator*

$$\langle W_\xi(\beta) \rangle_\xi \equiv \bar{W}(\beta) = \exp(\Delta(\beta)),$$

$$\bar{\rho}_Q(\beta) = \langle U_\xi(\beta) \rangle_\xi = e^{-\beta H_Q} e^{\Delta(\beta)}, \quad (48)$$

where the influence exponent $\Delta(\beta)$ is defined in accordance with Eq. (A12). It is convenient to perform a final massaging of this exponent to remove the explicit time ordering, the details of which we relegate to Appendix A. After this manipulation, $\Delta(\beta)$ takes the form

$$\Delta(\beta) \equiv \int_0^\beta d\tau \int_0^\tau d\tau' K(\tau - \tau') \tilde{f}(\tau) \tilde{f}(\tau'). \quad (49)$$

To progress, we introduce the adjoint chain generated by H_Q acting on the coupling,

$$f_n \equiv \text{ad}_{H_Q}^n(f), \quad \text{ad}_{H_Q}(A) \equiv [H_Q, A], \quad (50)$$

so that the interaction-picture operator has the exact series

$$\tilde{f}(\tau) = e^{\tau H_Q} f e^{-\tau H_Q} = \sum_{n \geq 0} \frac{\tau^n}{n!} f_n. \quad (51)$$

Substituting Eq. (51) into Eq. (49) yields

$$\Delta(\beta) = \sum_{n,m \geq 0} C_{nm}^>(\beta) f_n f_m, \quad (52)$$

where we have further defined an *ordered kernel-moment matrix*

$$C_{nm}^>(\beta) \equiv \int_0^\beta d\tau \int_0^\tau d\tau' K(\tau - \tau') \frac{\tau^n}{n!} \frac{\tau'^m}{m!}. \quad (53)$$

Here the notation $>$ is used to track the triangular domain of integration, where τ (associated with n) is the upper limit of integration for τ' (associated with m).

With this definition, all bath/temperature dependence enters $\Delta(\beta)$ only through the scalar moments $C_{nm}^>(\beta)$, while all operator structure is carried by the adjoint chain $\{f_n\}$. The final step is to re-express the averaged propagator in a *single* exponential. Using Eq. (48) we write

$$\bar{\rho}_Q(\beta) = e^{-\beta H_Q} e^{\Delta(\beta)} \equiv e^{-\beta H_{\text{MF}}(\beta)}, \quad (54)$$

where the mean-force Hamiltonian satisfies

$$-\beta H_{\text{MF}}(\beta) = \log \left(e^{-\beta H_Q} e^{\Delta(\beta)} \right). \quad (55)$$

Let $A \equiv -\beta H_Q$ and $B \equiv \Delta(\beta)$. The Baker–Campbell–Hausdorff (BCH) series [48–50] gives

$$\log(e^A e^B) = A + B + \frac{1}{2}[A, B] + \frac{1}{12}[A, [A, B]] + \frac{1}{12}[B, [B, A]] + \dots \quad (56)$$

Dividing by $-\beta$ and using $A = -\beta H_Q$ yields the expansion

$$H_{\text{MF}}(\beta) = H_Q - \frac{1}{\beta} \Delta(\beta) + \frac{1}{2}[H_Q, \Delta(\beta)] - \frac{\beta}{12}[H_Q, [H_Q, \Delta(\beta)]] + \frac{1}{12}[\Delta(\beta), [\Delta(\beta), H_Q]] + \dots \quad (57)$$

Expressed in this way, the closure content of $H_{\text{MF}}(\beta)$ is transparent: all terms are built from repeated commutators of H_Q acting on Δ , together with higher BCH terms involving commutators among those objects.

To see the explicit generator, introduce the adjoint superoperator $\text{ad}_{H_Q}(\cdot) \equiv [H_Q, \cdot]$. Then Eq. (57) becomes

$$H_{\text{MF}}(\beta) = H_Q - \frac{1}{\beta} \Delta + \frac{1}{2} \text{ad}_{H_Q} \Delta - \frac{\beta}{12} \text{ad}_{H_Q}^2 \Delta + \frac{1}{12} [\Delta, \text{ad}_{H_Q} \Delta] + \dots \quad (58)$$

Finally, because $\Delta(\beta)$ is quadratic in the adjoint chain, each commutator with H_Q simply shifts indices:

$$\text{ad}_{H_Q}(f_n f_m) = [H_Q, f_n f_m] = f_{n+1} f_m + f_n f_{m+1}. \quad (59)$$

Hence the first commutator term takes the explicit form

$$[H_Q, \Delta(\beta)] = \sum_{n,m \geq 0} C_{nm}^>(\beta) (f_{n+1} f_m + f_n f_{m+1}), \quad (60)$$

and higher $\text{ad}_{H_Q}^k \Delta$ generate the same family of products $f_a f_b$ with shifted indices. The remaining BCH terms (e.g. $[\Delta, \text{ad}_{H_Q} \Delta]$) introduce commutators among these quadratic elements, and are the sole source of additional operator growth beyond the quadratic span. The only remaining question is whether the operator family generated by Eq. (58) can be brought to a closed form.

The preceding identities imply that the entire BCH commutator tower *linear* in Δ remains within the quadratic span $\text{span}\{f_n f_m\}$. Terms to this order explicitly admit a closed resummation. Let $A \equiv -\beta H_Q$ and $B \equiv \Delta(\beta)$. Then the BCH logarithm satisfies the standard linear-in- B identity [51, 52]

$$\begin{aligned} \log(e^A e^B) &= A + \Phi(\text{ad}_A) B + O(B^2), \\ \Phi(x) &\equiv \frac{x}{1 - e^{-x}}. \end{aligned} \quad (61)$$

Substituting $A = -\beta H_Q$ and dividing by $-\beta$ gives the mean-force Hamiltonian to first order in Δ :

$$H_{\text{MF}}(\beta) = H_Q - \frac{1}{\beta} \Phi(\beta \text{ad}_{H_Q}) \Delta(\beta) + O(\Delta^2). \quad (62)$$

Expanding Φ yields the explicit commutator series with Bernoulli numbers B_k ,

$$\begin{aligned} H_{\text{MF}}(\beta) &= H_Q - \frac{1}{\beta} \sum_{k \geq 0} \frac{B_k}{k!} (-\beta)^k \text{ad}_{H_Q}^k \Delta(\beta) + O(\Delta^2), \\ \Phi(x) &= \sum_{k \geq 0} \frac{B_k}{k!} (-x)^k. \end{aligned} \quad (63)$$

Using Eq. (59) repeatedly, the iterated commutators admit the closed binomial form

$$\text{ad}_{H_Q}^k (f_n f_m) = \sum_{j=0}^k \binom{k}{j} f_{n+j} f_{m+k-j}, \quad (64)$$

so every term in Eq. (63) is explicitly a linear combination of products $f_a f_b$ with scalar coefficients determined solely by $C_{nm}^>(\beta)$ and universal combinatorics. Because of this, we may bring the first order contribution into a particularly compact form by introducing the discrete shift operator D . This will act on the moment matrix $C^>$ as

$$\begin{aligned} (DC^>)_{ab} &\equiv C_{a-1,b}^> + C_{a,b-1}^>, \\ C_{rs}^> &\equiv 0 \quad \text{if any index is negative.} \end{aligned} \quad (65)$$

Then Eq. (59) implies, after reindexing,

$$\text{ad}_{H_Q} \Delta(\beta) = \sum_{a,b \geq 0} (DC^>(\beta))_{ab} f_a f_b, \quad (66)$$

and by iteration,

$$\text{ad}_{H_Q}^k \Delta(\beta) = \sum_{a,b \geq 0} (D^k C^>(\beta))_{ab} f_a f_b. \quad (67)$$

Equivalently, $(D^k C^>)_{ab} = \sum_{j=0}^k \binom{k}{j} C_{a-j, b-(k-j)}^>$, which is the binomial index-shift structure implied by Eq. (64). From this we obtain the following form for $H_{\text{MF}}(\beta)$ to linear order in Δ :

$$\begin{aligned} H_{\text{MF}}(\beta) &= H_Q - \frac{1}{\beta} \sum_{a,b \geq 0} \tilde{C}_{ab}^>(\beta) f_a f_b \\ &\quad + O(\Delta^2), \end{aligned} \quad (68)$$

where the *renormalised moment matrix* $\tilde{C}^>(\beta)$ is the universal transform

$$\tilde{C}^>(\beta) \equiv \Phi(\beta D) C^>(\beta) = \sum_{k \geq 0} \frac{B_k}{k!} (-\beta)^k D^k C^>(\beta). \quad (69)$$

Thus, to linear order in the influence operator, the mean-force Hamiltonian is *exactly* a quadratic form in the products $f_a f_b$, with all bath/temperature dependence entering only through the scalar matrix $\tilde{C}_{ab}^>(\beta)$. Any operator growth beyond the quadratic span is confined to the nonlinear BCH sector $O(\Delta^2)$ (e.g. $[\Delta, \text{ad}_{H_Q} \Delta]$). While these terms are more difficult to handle as they involve commutators of quadratic forms, they nevertheless admit a precise series expansion via the full BCH formula Eq. (56). With this form in hand, we can now state a theorem for the representability of $H_{\text{MF}}(\beta)$ in a given operator family $\mathcal{A}$ as a *closure criterion*.

Theorem: Finite representability of mean force

Fix a target operator family $\mathcal{A}$ (e.g. k -local Pauli strings, a Lie algebra plus identity, or a finite-dimensional associative operator algebra) in which we seek to represent $H_{\text{MF}}(\beta)$. The Gaussian construction above allows us to formulate the exact criteria for representability as a closure theorem. In a physical sense, these conditions determine whether a valid *local generator* can be defined for the system at all; failure of closure implies that the effective Hamiltonian is fundamentally non-local with respect to the algebra $\mathcal{A}$.

Theorem (Closure of the Mean Force). *Let $\mathcal{A}$ be a target operator space containing the bare system Hamiltonian H_Q and coupling f . The Hamiltonian of Mean Force $H_{\text{MF}}(\beta)$ admits a finite representation within $\mathcal{A}$ if and only if the following three closure conditions are satisfied:*

(C1) *Adjoint Closure (No linear operator growth).* The subspace generated by repeated commutators of the coupling f with H_Q must remain inside $\mathcal{A}$:

$$\begin{aligned} f \in \mathcal{A}, \quad \text{ad}_{H_Q}(\mathcal{A}) &\subseteq \mathcal{A}, \\ \text{equivalently } \text{span}\{f_n\}_{n \geq 0} &\subseteq \mathcal{A}. \end{aligned} \quad (70)$$

(C2) *Quadratic Closure (No growth in the Gaussian sector).* Because the exact influence operator $\Delta(\beta)$ is

quadratic in the adjoint chain, the quadratic span generated by $\{f_n\}$ must also lie in $\mathcal{A}$:

$$f_n f_m \in \mathcal{A} \quad \text{for all indices that contribute in } \Delta(\beta), \quad (71)$$

so that $\Delta(\beta) \in \mathcal{A}$. A sufficient (and common) condition is that $\mathcal{A}$ is an associative algebra containing the identity and closed under multiplication.

(C3) *BCH Closure (No growth from commutators among quadratic elements)*. To ensure that the full BCH logarithm $\log(e^{-\beta H_Q} e^\Delta)$ remains in $\mathcal{A}$, the commutators generated among the quadratic elements must also close in $\mathcal{A}$; in particular

$$[\Delta(\beta), \text{ad}_{H_Q}^k \Delta(\beta)] \in \mathcal{A} \quad \text{for all } k \geq 0, \quad (72)$$

and similarly for the higher nested commutators appearing in the BCH series. A sufficient (and common) condition is again that $\mathcal{A}$ is a finite-dimensional associative algebra (or a Lie algebra containing Δ and closed under commutators), in which case all BCH commutators remain in $\mathcal{A}$ by construction.

Consequence (closed-form mean-force Hamiltonian). If (C1)–(C3) hold, then $\Delta(\beta) \in \mathcal{A}$ and all BCH commutators generated by Eq. (73) remain in $\mathcal{A}$. Since $H_Q \in \mathcal{A}$ by assumption, it follows that

$$\begin{aligned} \bar{\rho}_Q(\beta) &= e^{-\beta H_{\text{MF}}(\beta)}, \\ -\beta H_{\text{MF}}(\beta) &= \log\left(e^{-\beta H_Q} e^{\Delta(\beta)}\right), \end{aligned} \quad (73)$$

defines an $H_{\text{MF}}(\beta) \in \mathcal{A}$ and hence a closed-form representation of $H_{\text{MF}}(\beta)$ (up to the usual additive scalar fixed by Z_X). If either (C1) fails (adjoint-chain growth) or (C2)–(C3) fail (growth in the quadratic/BCH sector), then $\Delta(\beta)$ and/or the BCH commutator tower generates operators outside $\mathcal{A}$; any representation within $\mathcal{A}$ is necessarily truncated or projected.

This statement leads to an important corollary. For continuous variable systems $H_Q = p^2/2m + V(x)$, if the potential $V(x)$ is a polynomial of degree greater than 2, the adjoint chain $\{f_n\}$ generally does not close in any finite-dimensional Lie algebra; instead, it generates the full universal enveloping algebra. Consequently, no finite representation of the mean force exists within the original operator class. This obstruction is analogous (in spirit) to the Groenewold–Van Hove no-go theorem on consistent quantisation of polynomial observables: the failure to close the Lie algebra under adjoint action corresponds to the impossibility of mapping the quantum evolution of non-quadratic potentials to a finite phase space flow. For such potentials, there is no *local operator* that exactly describes the reduced equilibrium state.

From a practical standpoint, the closure of $\mathcal{A}$ is not necessary for an arbitrarily accurate approximation of $H_{\text{MF}}(\beta)$, as the BCH logarithm $\log(e^{-\beta H_Q} e^\Delta)$ still yields a controlled expansion. The operator content is generated by the adjoint chain and its quadratic products, while all bath dependence remains scalar through

$C_{nm}^{>}(\beta)$. One may therefore truncate (in commutator depth, locality class, or operator weight) without introducing any approximation on the bath side.

This algebraic perspective also clarifies approximate open-system methods. Numerical schemes such as the polaron transformation [25, 47], hierarchical equations of motion [32], and chain-mapping/DMRG techniques [34] may be understood as distinct choices of the target operator family $\mathcal{A}$ or distinct rules for enlarging it. The polaron frame dresses $\mathcal{A}$ with coherent displacements; HEOM truncates the memory depth of the influence functional while mirroring operator growth in the adjoint chain; and chain mappings truncate the spatial extent of the harmonic bath. The closure problem is therefore equivalent to identifying a reduced model class that is large enough to contain the logarithm of the quenched propagator, yet small enough to remain usable.

Constructive reduced generators and compact model classes

At the level of the global Gibbs operator, the equilibrium problem is linear: one exponentiates H_{tot} and traces over the bath. The difficulty appears only after reduction. The partial trace removes environmental degrees of freedom explicitly, but the subsystem generator is recovered only after the nonlinear compression $H_{\text{MF}}(\beta) = -(1/\beta) \log \bar{\rho}_Q(\beta)$. The HMF is therefore not merely another name for the reduced state. It is the constructive object that tells us whether the reduced description can still be written inside a tractable operator family.

Viewed this way, Conditions (C1)–(C3) are model-class conditions. Adjoint closure asks whether repeated system–bath back-action remains inside the chosen feature set. Quadratic closure asks whether the Gaussian influence can be expressed in that same set. BCH closure asks whether recombining the reduced state with the bare Gibbs factor introduces genuinely new operators or stays compact. When all three hold, the reduced equilibrium admits an explicit generator that is exact, finite, and inspectable.

The failure modes are equally informative. If the adjoint chain proliferates without closure, then the reduced generator cannot be compressed into the proposed ansatz without loss. This is the operator-level version of a representability bottleneck: the hidden environment has injected structure that the target model class cannot faithfully contain. Approximate methods then differ less by the bath data they use than by the restricted operator families they choose to keep.

This language has a natural, but only heuristic, relation to current machine-learning questions. Mechanistic interpretability asks when the action of a large system can be summarized by a smaller explicit mechanism.

Here the target is not a trained network but the reduced equilibrium operator of an open quantum system. The mathematics is different, but the structural question is similar: when does a large hidden substrate admit a compact generator, and when does feature growth force one outside any simple interpretable class?

SPIN-BOSON MODEL AS A CLOSED REDUCED GENERATOR

We now instantiate the closure logic of Secs. and in the spin-boson model [19, 47]. This is the minimal noncommuting testbed: the system algebra is just rich enough to generate nontrivial reduced structure, yet small enough that the generator can be obtained exactly. The point is not that qubits are unusually simple; it is that here the closure criterion can be followed all the way to a closed reduced generator.

Let us first set conventions. During derivation we work in the qubit energy basis, $\hat{\mathbf{n}}_s = \hat{\mathbf{z}}$, and then rewrite the final answer covariantly in the coupling plane spanned by $\hat{\mathbf{n}}_s$ and $\hat{\mathbf{f}}$. We take

$$H_Q = \frac{\omega_q}{2} \hat{\mathbf{n}}_s \cdot \boldsymbol{\sigma}, \quad f = \hat{\mathbf{f}} \cdot \boldsymbol{\sigma}, \quad (74)$$

with $\hat{\mathbf{f}} = (s \cos \phi, s \sin \phi, c)$, $c \equiv \cos \theta$, $s \equiv \sin \theta$. The overall coupling amplitude g is suppressed in intermediate expressions and reintroduced when discussing scaling.

Bath kernels and the symmetrized influence state

The interaction-picture coupling operator is

$$\tilde{f}(\tau) = e^{\tau H_Q} f e^{-\tau H_Q} = c \sigma_z + f_- e^{\omega_q \tau} \sigma_+ + f_+ e^{-\omega_q \tau} \sigma_-, \quad (75)$$

with $f_{\pm} = s e^{\pm i \phi}$. The Pauli algebra then implies that all bath dependence enters through two real response channels,

$$\begin{aligned} \Sigma_z &\equiv \frac{1}{2} [R(\omega_q) - R(-\omega_q)], \\ \Sigma_{\perp} &\equiv \frac{4}{\omega_q} \left[\cosh\left(\frac{\beta \omega_q}{2}\right) \mathcal{K}(0) - e^{-\beta \omega_q/2} \mathcal{K}(\omega_q) \right]. \end{aligned} \quad (76)$$

The reduced operator can be written in the symmetrized form $\bar{\rho}_Q = \Pi^{1/2} e^S \Pi^{1/2}$, with $\Pi = e^{-\beta H_Q}$ and

$$S = \Delta_0 \mathbb{I} + s^2 \Sigma_z \sigma_z - s c \Sigma_{\perp} (\cos \phi \sigma_x + \sin \phi \sigma_y). \quad (77)$$

Subtracting the scalar piece, $M \equiv S - \Delta_0 \mathbb{I}$, gives an influence Bloch vector

$$\mathbf{m}_S = s \Sigma_{\perp} \hat{\mathbf{f}}_{\perp} + s^2 (\Sigma_z - \Sigma_{\perp}) \hat{\mathbf{n}}_s, \quad (78)$$

where $\hat{\mathbf{f}}_{\perp} = (-c \cos \phi, -c \sin \phi, s)$ lies in the coupling plane and is orthogonal to $\hat{\mathbf{f}}$.

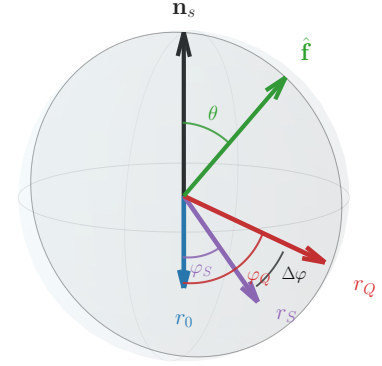


FIG. 2. Bloch-sphere geometry for the exact qubit construction. The bare thermal state has radius r_0 (blue), the symmetrized influence state $\rho_S \propto e^S$ has radius r_S and tilt φ_S (purple), and the final reduced state has radius r_Q and tilt φ_Q (red). The exact reduced generator stays in the same coupling plane throughout.

The corresponding influence magnitude is

$$\chi = \|\mathbf{m}_S\| = |s| \sqrt{c^2 \Sigma_{\perp}^2 + s^2 \Sigma_z^2}, \quad (79)$$

and the normalized influence state is

$$\rho_S = \frac{e^S}{\text{Tr}(e^S)} = \frac{1}{2} [\mathbb{I} + \tanh \chi \hat{\mathbf{m}}_S \cdot \boldsymbol{\sigma}], \quad \hat{\mathbf{m}}_S = \mathbf{m}_S / \chi. \quad (80)$$

Its tilt inside the coupling plane is fixed by the ratio of response channels,

$$\tan \varphi_S = -\frac{c}{s} \frac{\Sigma_{\perp}}{\Sigma_z}. \quad (81)$$

At this stage the constructive content is already visible: the entire bath has compressed to two scalar channels and one influence magnitude.

Closed reduced state and generator

Recombining the influence state with the bare Gibbs factor yields the reduced state

$$\rho_Q = \frac{1}{2} [\mathbb{I} + r_Q \hat{\mathbf{m}}_Q \cdot \boldsymbol{\sigma}], \quad (82)$$

with Bloch components

$$m_z = r_Q \cos \varphi_Q, \quad m_{\perp} = r_Q \sin \varphi_Q. \quad (83)$$

The longitudinal scale is controlled by

$$\Theta \equiv \operatorname{arctanh}(\gamma(\chi) s^2 \Sigma_z) - \frac{\beta \omega_q}{2}, \quad (84)$$

$$\gamma(\chi) \equiv \tanh \chi / \chi, \quad (85)$$

which gives the exact solution

$$m_z = \tanh \Theta, \quad (86)$$

$$m_\perp = \tan \varphi_S \left(m_z \cosh \frac{\beta \omega_q}{2} + \sinh \frac{\beta \omega_q}{2} \right). \quad (87)$$

The final tilt and radius follow as

$$\tan \varphi_Q = \tan \varphi_S \frac{m_z + r_0}{\sqrt{1 - r_0^2} m_z}, \quad r_0 = \tanh \left(\frac{\beta \omega_q}{2} \right), \quad (88)$$

and

$$1 - r_Q^2 = \frac{(1 - r_0^2)(1 - r_S^2)}{(1 + r_0 r_S \cos \varphi_S)^2}, \quad r_S = \tanh \chi. \quad (89)$$

For any full-rank qubit state, the operator logarithm is elementary. Writing $\hat{\mathbf{m}}_Q = (m_\perp \cos \phi, m_\perp \sin \phi, m_z)/r_Q$, the exact qubit Hamiltonian of mean force is therefore

$$H_{\text{MF}}(\beta) = h_0(\beta) \mathbb{I} - \frac{1}{\beta} \operatorname{arctanh}(r_Q) \hat{\mathbf{m}}_Q \cdot \boldsymbol{\sigma}, \quad (90)$$

where the scalar $h_0(\beta)$ fixes normalization and carries the additive identity freedom of the HMF. Equation (90) makes the closure statement concrete: the exact reduced generator remains inside the same Bloch-plane operator family throughout, while the bath enters only through the two response channels $(\Sigma_z, \Sigma_\perp)$ and the influence scale χ .

EXACT-THEORY DIAGNOSTICS AND THE RESPONSE SCALE

The exact qubit solution isolates a single scalar control variable, χ , so it is natural to organize diagnostics around it. Reintroducing an explicit dimensionless coupling strength through

$$K(u) = g^2 K_0(u), \quad (91)$$

all response kernels inherit the same quadratic scaling and the influence magnitude factorizes as

$$\chi(\beta, g, \theta) = g^2 \chi_0(\beta, \theta). \quad (92)$$

The entire recombination of bare Gibbs weight and bath influence is therefore controlled by the universal scalar function

$$\gamma(\chi) = \frac{\tanh \chi}{\chi}. \quad (93)$$

The exact crossover scale is determined by $\chi \sim 1$, or equivalently

$$\chi(\beta, g, \theta) \sim 1 \iff g^2 \sim \chi_0(\beta, \theta)^{-1}, \quad (94)$$

which defines the coupling scale

$$g_\star(\beta) = \chi_0(\beta, \theta)^{-1/2}. \quad (95)$$

No approximation has been made here: $g_\star(\beta)$ is an exact property of the reduced generator.

Two benchmark asymptotes remain useful. For $\chi \ll 1$,

$$\gamma(\chi) \simeq 1 - \frac{\chi^2}{3}, \quad (96)$$

so the reduced state tracks the bare Gibbs description with perturbative bath-induced corrections. For $\chi \gg 1$,

$$\gamma(\chi) \simeq \chi^{-1}, \quad (97)$$

so the reduced generator is dominated by the normalized influence direction. These asymptotes are best read as validation benchmarks, not as the organizing story of the exact solution. The main point is that the full crossover is captured by one response coordinate.

The same coordinate controls observable diagnostics. Differentiating the exact Bloch components shows that the angle susceptibility and radius gradient peak at fixed values of χ , namely $\chi_{\text{peak},\varphi} \approx 0.42$ and $\chi_{\text{peak},r} \approx 0.85$ for the Ohmic example used below. After rescaling the coupling as $g/g_\star(\beta)$, these peaks align across temperatures. The alignment is not a numerical accident; it is the statement that the response scale is intrinsic to the exact reduced generator.

To make these statements concrete we choose the windowed Ohmic spectral density

$$J(\omega) = g^2 \omega e^{-\omega/\omega_c}, \quad \omega \in [0, 2\omega_q], \quad (98)$$

with cutoff $\omega_c = 5\omega_q$. The ultraviolet window matches the finite-mode simulations of the next section, while the exponential cutoff regularizes the high-frequency tail [53]. Having fixed the bath, we set $\theta = \pi/4$ and $\omega_q = 1$, since the remaining behavior is already captured by the universal χ -scaling.

REPRESENTABILITY DIAGNOSTICS IN TRUNCATED BATHS

The exact qubit solution of Sec. depends on the bath only through response data at three frequencies, $\omega \in \{0, \pm\omega_q\}$. To test it we compare against exact diagonalization (ED) of a truncated spin-boson model, where the qubit is coupled to N_ω harmonic oscillators and each mode is represented with a finite Fock cutoff n_{max} . A useful feature of this comparison is that the analytic kernel integrals can be evaluated in two ways: using the

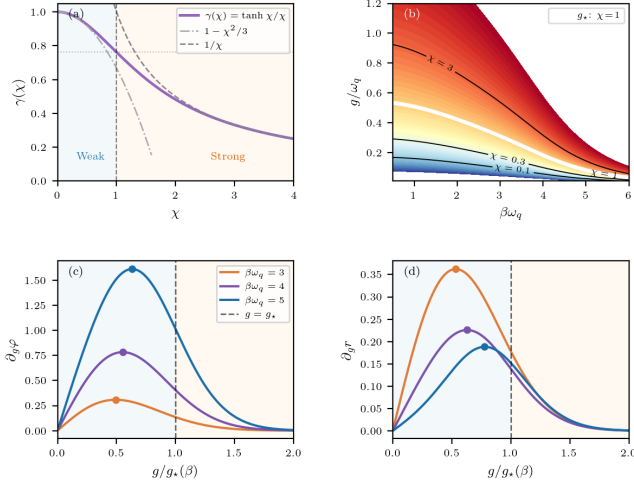


FIG. 3. Exact diagnostics organized by the response scale $\chi(\beta, g) = g^2 \chi_0(\beta)$ for the Ohmic bath (98) at $\theta = \pi/4$. Panel (a): Universal recombination function $\gamma(\chi) = \tanh \chi / \chi$ together with its small- χ and large- χ benchmark asymptotes. Panel (b): Map of $\chi(\beta, g)$ showing the exact crossover scale $g_*(\beta) = \chi_0(\beta)^{-1/2}$. Panels (c,d): Angle and radius diagnostics plotted against the scaled coupling $g/g_*(\beta)$, which aligns the peak positions across temperatures.

full continuum spectral density (the *continuous analytic* branch) or using the same discrete mode set seen by ED (the *discrete analytic* branch).

Figure 4 compares the excited-state population p_{11} across coupling strengths and inverse temperatures for $N_\omega = 40$ modes. At small χ , all three curves coincide. As the system crosses the exact response scale $\chi \sim 1$, the analytic predictions separate from ED. The key point is that the discrete analytic branch does *not* follow the numerical deviation: it bends back toward the continuum benchmark. This shows that the exact HMF theory is already correct for the discrete mode set itself. The discrepancy is therefore not a failure of the theory, nor a finite-mode artifact.

The remaining gap is a representability bottleneck in the truncated Hilbert space. In the large-response regime the coupled system approaches a displaced, polaron-like ground state, and faithfully representing that state requires $n_{\max} \gtrsim |\alpha_k|^2$ for each displaced mode $\alpha_k \sim g_k/\omega_k$. Once $\chi \gtrsim 1$, the chosen Fock cutoff can no longer support the required occupations. ED is then trapped in a model class that is too small to contain the exact dressed equilibrium state.

The same interpretation explains the bandwidth sensitivity in Fig. 5. Varying the spectral window changes how the displacement load is distributed over the available modes. As the system is pushed across the response scale, the required occupations grow and the available Fock space is exhausted. The sharp peak in cutoff sensitivity is therefore a direct signature of the finite basis

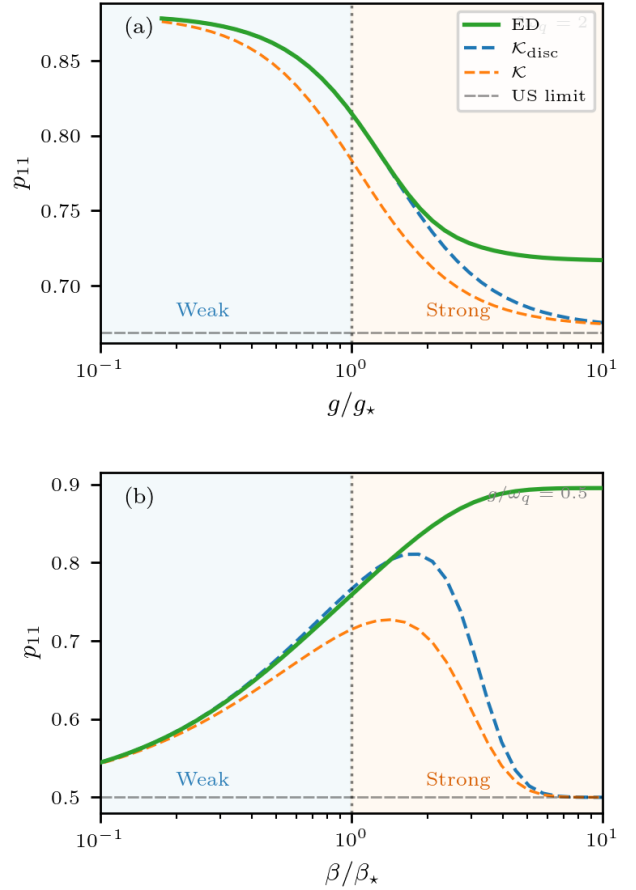


FIG. 4. Excited-state population p_{11} across the exact response scale for $N_\omega = 40$ modes. Green solid: ED. Blue dashed: discrete analytic theory using the same mode set. Orange dotted: continuum analytic theory. Dashed horizontal lines mark the large-response benchmark p_∞ . The gap between ED and the analytic branches is a representability failure of the truncated basis, not a breakdown of the exact reduced generator.

hitting its representational boundary.

The numerical message is therefore narrower, and stronger, than a generic “agreement with simulation” claim. The analytic HMF provides the correct reduced target for both discrete and continuous baths. The observed numerical deviations diagnose when a chosen finite model class is no longer expressive enough to realize that target.

DISCUSSION

This paper makes three connected points. First, the Hamiltonian of mean force is best viewed as a constructive reduced generator, not merely as a formal logarithm. Second, in the Gaussian sector this constructive question reduces to a closure problem for the adjoint chain gener-

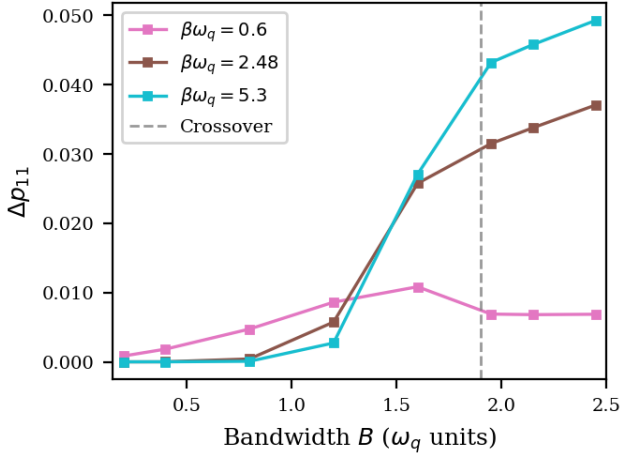


FIG. 5. Cutoff sensitivity $\Delta p_{11} \equiv |p_{11}(n_{\max} = 4) - p_{11}(n_{\max} = 6)|$ as a function of spectral bandwidth B in units of ω_q . The sensitivity peak marks the onset of the representability bottleneck: beyond this point the truncated basis can no longer accommodate the mode occupations required by the exact dressed state.

ated by H_Q and the coupling operator. Third, the same closure logic explains both exact solvability and the failure modes of finite-basis numerics.

Closure and constructive reduced generators

The central algebraic result is that representability of the HMF inside a restricted operator family has an exact answer. Conditions (C1)–(C3) separate the problem into three layers: whether repeated adjoint action grows the operator basis, whether the Gaussian influence remains inside that basis, and whether the BCH recombination generates further operators. When all three hold, the reduced equilibrium admits an exact finite generator. When they fail, the failure is structural rather than computational: the reduced state can still be computed, but not compressed losslessly into the proposed ansatz.

This makes the closure theorem useful beyond the present example. It turns the usual question “can we evaluate the reduced state?” into the sharper question “which operator family is large enough to contain the exact reduced generator?” In that form, the theorem is already a guide to controlled approximation. One may truncate the operator side without approximating the bath side, and the resulting error has a clear interpretation as model-class error.

The qubit as exact operator compression

For the spin-boson qubit, closure is exact. The reduced generator remains in the Bloch plane determined

by the system axis and the coupling axis, and the entire bath compresses to two scalar response channels together with the influence magnitude χ . This is a particularly transparent example of constructive reduction: a large environment is integrated out, yet the subsystem retains a closed generator with explicit geometry and explicit dependence on the bath response.

The qubit result is valuable not because qubits are the final target, but because they show the mechanism without approximation. The exact generator does not emerge from summing a perturbative series term by term. It emerges because, after the bath trace is performed, the remaining operator algebra closes. That distinction matters. It identifies what should generalize to larger systems: not the specific Pauli formulas, but the separation between bath data and operator growth.

Representability bottlenecks and finite-resource inference

The numerical diagnostics sharpen the interpretation. ED disagrees with the analytic prediction not when the theory becomes inaccurate, but when the chosen finite basis is no longer expressive enough to realize the exact dressed state. The response scale $\chi \sim 1$ marks the onset of this bottleneck in the examples studied here. In other words, the crossover is simultaneously physical and representational: it tracks both the reorganization of the reduced state and the point at which a restricted model class begins to fail.

This is why the distinction between discrete analytic and numerical ED matters. Once the discrete analytic branch peels away from ED and returns toward the continuum benchmark, the problem is no longer one of physics missed by the generator. It is one of structure missed by the basis. That diagnosis is useful well beyond this example, because many approximate methods in open-system physics are precisely choices of tractable operator or Hilbert-space classes.

Connections beyond open-system thermodynamics

Recent work on physical theories of representation emphasizes that exact state descriptions are less useful than explicit generators that expose what must be retained under coarse-graining [1]. The HMF offers a concrete quantum realization of that principle. The partial trace hides a large environment inside a reduced state; closure determines whether that hidden structure can still be recast as a compact generator.

This is also where the analogy to machine learning becomes useful. The information-bottleneck viewpoint asks when a compressed variable preserves the relevant structure of a larger system [2]. Mechanistic interpretability

asks when a complicated hidden process can be summarized by a smaller explicit mechanism. The present work does not claim an identity with either framework. It does, however, provide a precise equilibrium problem with the same structural flavor: a large hidden substrate is reduced, and the main question is whether the result stays inside a compact, interpretable model class.

Limitations and outlook

The main analytic restriction is Gaussianity. The quenched representation itself is more general, but the exact collapse of the influence to a bilinear operator depends on the bath cumulant hierarchy truncating at second order. Beyond that sector, higher cumulants generate higher operator monomials and the closure problem becomes correspondingly richer.

Within the Gaussian sector, the next natural targets are multi-qubit systems and other finite-dimensional models where the adjoint chain generates nontrivial many-body operator content. There the central question is no longer whether an HMF exists, but how rapidly operator growth escapes a chosen locality class. In that sense the present qubit solution is a starting point for a broader program of constructive reduced generators in interacting systems.

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Appendix A: Averaging of the Time-Ordered Gaussian Propagator

We provide the explicit derivation of the identity in Eq. (A12), which relates the expectation value of a time-ordered exponential driven by Gaussian noise to a bilocal influence phase. Consider the time-ordered propagator

$$W_\xi(\beta) = \mathcal{T}_\tau \exp \left[- \int_0^\beta d\tau \xi(\tau) \tilde{f}(\tau) \right], \quad (\text{A1})$$

where $\xi(\tau)$ is a centered Gaussian stochastic field with covariance $\langle \xi(\tau) \xi(\tau') \rangle = K(\tau - \tau')$, and $\tilde{f}(\tau)$ is a system operator. We wish to prove that

$$\langle W_\xi(\beta) \rangle_\xi = \exp \left[\frac{1}{2} \int_0^\beta d\tau \int_0^\beta d\tau' K(\tau - \tau') \times \mathcal{T}_\tau \left[\tilde{f}(\tau) \tilde{f}(\tau') \right] \right]. \quad (\text{A2})$$

The Dyson expansion of the time-ordered exponential is given by

$$W_\xi(\beta) = \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} \int_0^\beta d\tau_1 \cdots d\tau_n \times \mathcal{T}_\tau \left[\xi(\tau_1) \tilde{f}(\tau_1) \cdots \xi(\tau_n) \tilde{f}(\tau_n) \right]. \quad (\text{A3})$$

Because each $\xi(\tau_i)$ is a scalar (c-number), the time-ordering operator acts only on the operator factors. Thus, we may pull the stochastic fields out of the ordering:

$$\mathcal{T}_\tau \left[\xi_1 \tilde{f}_1 \cdots \xi_n \tilde{f}_n \right] = (\xi_1 \cdots \xi_n) \mathcal{T}_\tau \left[\tilde{f}_1 \cdots \tilde{f}_n \right], \quad (\text{A4})$$

where $\xi_i \equiv \xi(\tau_i)$ and $\tilde{f}_i \equiv \tilde{f}(\tau_i)$. Taking the stochastic average over ξ and noting that centered Gaussian noise has vanishing odd moments, the expansion becomes

$$\langle W_\xi \rangle = \sum_{p=0}^{\infty} \frac{1}{(2p)!} \int d^{2p}\tau \langle \xi_1 \cdots \xi_{2p} \rangle \mathcal{T}_\tau \left[\tilde{f}_1 \cdots \tilde{f}_{2p} \right]. \quad (\text{A5})$$

By Wick's theorem, the even moments are given by the sum over all pairings P :

$$\langle \xi_1 \cdots \xi_{2p} \rangle = \sum_P \prod_{(i,j) \in P} K(\tau_i - \tau_j). \quad (\text{A6})$$

We now consider the expansion of the claimed exponential result. Define the bilocal operator

$$X \equiv \frac{1}{2} \int_0^\beta d\tau \int_0^\beta d\tau' K(\tau - \tau') \mathcal{T}_\tau \left[\tilde{f}(\tau) \tilde{f}(\tau') \right]. \quad (\text{A7})$$

The p -th term in the series for e^X is $(1/p!)X^p$. Introducing p independent pairs of integration variables (τ_{2r-1}, τ_{2r}) , we write

$$X^p = \left(\frac{1}{2} \right)^p \int d^{2p}\tau \prod_{r=1}^p \left[K_{(2r-1)(2r)} \mathcal{T}_\tau \left[\tilde{f}_{2r-1} \tilde{f}_{2r} \right] \right], \quad (\text{A8})$$

so that the full series is

$$e^X = \sum_{p=0}^{\infty} \frac{1}{p!} \left(\frac{1}{2} \right)^p \int d^{2p}\tau \left[\prod_{r=1}^p K_{(2r-1)(2r)} \right] \times \left[\prod_{r=1}^p \mathcal{T}_\tau \left[\tilde{f}_{2r-1} \tilde{f}_{2r} \right] \right]. \quad (\text{A9})$$

To relate this to Eq. (A5), we exploit the property of time ordering on the hypercube $[0, \beta]^{2p}$. apart from a set of measure zero, the integration domain can be partitioned into $(2p)!$ disjoint regions R_π defined by the strict total ordering $\tau_{\pi(1)} > \tau_{\pi(2)} > \dots > \tau_{\pi(2p)}$. On each such region, the global time ordering satisfies $\mathcal{T}_\tau[\tilde{f}_1 \dots \tilde{f}_{2p}] = \tilde{f}_{\pi(1)} \dots \tilde{f}_{\pi(2p)}$. Similarly, each pairwise time-ordering $\mathcal{T}_\tau[\tilde{f}_{2r-1} \tilde{f}_{2r}]$ collapses to the correctly oriented product according to the same strict ordering. It follows that the product of pair-ordered operators coincides with the global ordering almost everywhere:

$$\prod_{r=1}^p \mathcal{T}_\tau [\tilde{f}_{2r-1} \tilde{f}_{2r}] = \mathcal{T}_\tau [\tilde{f}_1 \dots \tilde{f}_{2p}]. \quad (\text{A10})$$

Substituting this into Eq. (A9), we may pull the global time ordering out of the products.

The final step requires counting the occurrences of each Wick pairing. The expansion of X^p utilizes p labelled slots $r = 1, \dots, p$, each containing an ordered pair. A Wick contraction P is a set of p unordered pairs. For any fixed Wick pairing P , there are $p!$ ways to assign the pairs to the p slots and 2^p ways to orient each pair. Each Wick pairing is therefore represented exactly $2^p p!$ times in the integrand. This multiplicity is canceled by the prefactor $(1/p!)(1/2)^p$, yielding

$$e^X = \sum_{p=0}^{\infty} \frac{1}{(2p)!} \int d^{2p} \tau \left(\sum_P \prod_{(i,j) \in P} K_{ij} \right) \mathcal{T}_\tau [\tilde{f}_1 \dots \tilde{f}_{2p}]. \quad (\text{A11})$$

This series matches Eq. (A5) term-by-term, proving the exponentiation of the Gaussian average. Consequently, we may right the influence on the reduced density matrix as

$$\langle W_\xi(\beta) \rangle_\xi = \exp \left[\frac{1}{2} \int_0^\beta d\tau \int_0^\beta d\tau' K(\tau - \tau') \times \mathcal{T}_\tau [\tilde{f}(\tau) \tilde{f}(\tau')] \right]. \quad (\text{A12})$$

Domain Folding and the Triangular Form of the Influence Operator

The influence operator $\Delta(\beta)$

$$\Delta(\beta) = \frac{1}{2} \int_0^\beta d\tau \int_0^\beta d\tau' K(\tau - \tau') \mathcal{T}_\tau [\tilde{f}(\tau) \tilde{f}(\tau')]. \quad (\text{A13})$$

admits a simplified representation over a strictly ordered time domain. By definition, the time-ordering operator for two system operators acts as

$$\mathcal{T}_\tau [\tilde{f}(\tau) \tilde{f}(\tau')] = \Theta(\tau - \tau') \tilde{f}(\tau) \tilde{f}(\tau') + \Theta(\tau' - \tau) \tilde{f}(\tau') \tilde{f}(\tau). \quad (\text{A14})$$

Substituting this into the integral and splitting the domain yields

$$\Delta(\beta) = \frac{1}{2} \iint_{\tau > \tau'} d\tau d\tau' K(\tau - \tau') \tilde{f}(\tau) \tilde{f}(\tau') + \frac{1}{2} \iint_{\tau' > \tau} d\tau d\tau' K(\tau - \tau') \tilde{f}(\tau') \tilde{f}(\tau). \quad (\text{A15})$$

In the second term, we perform the swap of dummy variables $(\tau, \tau') \mapsto (\tau', \tau)$. The region $\tau' > \tau$ becomes $\tau > \tau'$, the operator product transforms to $\tilde{f}(\tau) \tilde{f}(\tau')$, and the kernel becomes $K(\tau' - \tau)$. Thus,

$$\frac{1}{2} \iint_{\tau' > \tau} d\tau d\tau' K(\tau - \tau') \tilde{f}(\tau') \tilde{f}(\tau) = \frac{1}{2} \iint_{\tau > \tau'} d\tau d\tau' K(\tau' - \tau) \tilde{f}(\tau) \tilde{f}(\tau'). \quad (\text{A16})$$

Using the symmetry of the equilibrium kernel, $K(\tau - \tau') = K(\tau' - \tau)$, we find that both contributions are identical on the ordered region. Summing them yields the triangular representation used throughout this work:

$$\Delta(\beta) = \int_0^\beta d\tau \int_0^\tau d\tau' K(\tau - \tau') \tilde{f}(\tau) \tilde{f}(\tau'). \quad (\text{A17})$$

Appendix B: Algebraic Construction in the Pauli Basis

In the main text (Sec.), the exact solution was derived using the ladder basis $\sigma_\pm$, which diagonalises the imaginary-time evolution and leads directly to the response kernels. Here we afford an alternative perspective by working directly in the Pauli basis $\{\sigma_x, \sigma_y, \sigma_z\}$. While algebraically heavier, this route makes the operator closure explicit at the level of structure constants and facilitates the systematic summation of the BCH series.

1. Polylogarithmic decomposition from the adjoint chain

We consider the same transverse coupling model defined by

$$H_Q = \frac{\omega_q}{2} \sigma_z, \quad f = c \sigma_z - s \sigma_x, \quad (\text{B1})$$

with $c = \cos \theta$, $s = \sin \theta$. The adjoint chain $f_n \equiv \text{ad}_{H_Q}^n(f)$ is given by

$$f_n = \mathbf{f}_n \cdot \boldsymbol{\sigma}, \quad \mathbf{f}_n = \begin{cases} (-s, 0, c) & n = 0, \\ (-s\omega_q^n, 0, 0) & n \geq 1, n \text{ even}, \\ (0, -is\omega_q^n, 0) & n \geq 1, n \text{ odd}. \end{cases} \quad (\text{B2})$$

The recurrence ensures that $f_{n \geq 1}$ alternates between the x and y axes, while f_0 contains a persistent z -component.

The influence exponent $\Delta(\beta) = \sum_{n,m} C_{nm}^> f_n f_m$ is evaluated using the standard Pauli product rule $(\mathbf{a} \cdot \boldsymbol{\sigma})(\mathbf{b} \cdot \boldsymbol{\sigma}) = (\mathbf{a} \cdot \mathbf{b})\mathbb{I} + i(\mathbf{a} \times \mathbf{b}) \cdot \boldsymbol{\sigma}$. Decomposition of the ordered moments into symmetric (S_{nm}) and antisymmetric (A_{nm}) parts reveals that only the commutator sector A_{nm} contributes to the operator structure:

$$\Delta(\beta) \cong 2i \sum_{n>m} A_{nm} (\mathbf{f}_n \times \mathbf{f}_m) \cdot \boldsymbol{\sigma}, \quad (\text{B3})$$

up to a scalar shift $\Delta_0\mathbb{I}$. The cross products exhibit a strict parity selection rule:

- For $n, m \geq 1$: $\mathbf{f}_n \times \mathbf{f}_m$ lies along $\hat{\mathbf{z}}$ since both vectors are confined to the xy -plane.
- For $m = 0$: $\mathbf{f}_n \times \mathbf{f}_0$ generates terms in the xy -plane, mixing the longitudinal and transverse directions.

Explicitly summing these series yields the channel decomposition

$$\Delta = \Delta_0\mathbb{I} + \Delta_x\sigma_x + \Delta_y\sigma_y + \Delta_z\sigma_z, \quad (\text{B4})$$

where the coefficients are polylogarithmic series in ω_q :

$$\Delta_x = -2cs \sum_{\ell \geq 0} A_{2\ell+1,0}(\beta) \omega_q^{2\ell+1}, \quad (\text{B5})$$

$$\Delta_y = -2ics \sum_{\ell \geq 1} A_{2\ell,0}(\beta) \omega_q^{2\ell}, \quad (\text{B6})$$

$$\begin{aligned} \Delta_z = & -2s^2 \sum_{k \geq 1, \ell \geq 0} A_{2k,2\ell+1}(\beta) \omega_q^{2k+2\ell+1} \\ & - 2s^2 \sum_{\ell \geq 0} A_{2\ell+1,0}(\beta) \omega_q^{2\ell+1}. \end{aligned} \quad (\text{B7})$$

These series are the Pauli-basis equivalent of the closed-form expressions obtained in Sec. . The connection is established by identifying the sums as Taylor expansions of the kernels $\mathcal{G}^>(0, \pm\omega_q)$ and $\mathcal{G}^>(\pm\omega_q, \mp\omega_q)$.

2. Exact BCH resummation via the nilpotency identity

The central challenge in the mean-force construction is the evaluation of the BCH logarithm $-\beta H_{\text{MF}} = \log(e^{-\beta H_Q} e^\Delta)$. While the closure theorem guarantees that this remains within the algebra, a manual summation of the BCH series is laborious. Here we show how the $\mathfrak{su}(2)$ structure allows for an exact, non-perturbative resummation.

Let $M \equiv \Delta - \Delta_0\mathbb{I} = \boldsymbol{\Delta} \cdot \boldsymbol{\sigma}$ be the traceless part of the influence operator. The square of this operator satisfies a crucial identity:

$$M^2 = (\boldsymbol{\Delta} \cdot \boldsymbol{\sigma})^2 = (\boldsymbol{\Delta} \cdot \boldsymbol{\Delta})\mathbb{I} \equiv \chi^2\mathbb{I}, \quad (\text{B8})$$

where $\chi^2 = \Delta_x^2 + \Delta_y^2 + \Delta_z^2$ (or $\Delta_z^2 + \Sigma_+ \Sigma_-$ in the ladder gauge). This “generalized nilpotency” condition truncates the power series for the exponential e^M into even and odd sectors, yielding the exact Euler-like formula:

$$e^\Delta = e^{\Delta_0} \left[\cosh \chi \mathbb{I} + \frac{\sinh \chi}{\chi} M \right]. \quad (\text{B9})$$

This expression sums the internal structure of the influence operator potential to all orders. The full reduced state is then the product of two $\text{SL}(2, \mathbb{C})$ matrices:

$$\begin{aligned} \bar{\rho}_Q &= e^{-\beta H_Q} e^\Delta \\ &= e^{\Delta_0} \begin{pmatrix} e^{-\beta\omega_q/2} & 0 \\ 0 & e^{\beta\omega_q/2} \end{pmatrix} \begin{pmatrix} \cosh \chi + \varphi \Delta_z & \varphi(\Delta_x - i\Delta_y) \\ \varphi(\Delta_x + i\Delta_y) & \cosh \chi \end{pmatrix}, \end{aligned} \quad (\text{B10})$$

with $\varphi = \sinh \chi / \chi$. Since the product of 2×2 matrices is a 2×2 matrix, the BCH series (which is the logarithm of this product) is strictly confined to the Pauli algebra.

Explicitly, for a resulting state $\rho_Q = \frac{1}{2}(\mathbb{I} + \mathbf{r} \cdot \boldsymbol{\sigma})$, the mean-force Hamiltonian is

$$H_{\text{MF}} = c_0\mathbb{I} - \frac{1}{\beta} \frac{\text{arctanh } r}{r} \mathbf{r} \cdot \boldsymbol{\sigma}. \quad (\text{B11})$$

This formula represents the analytic summation of the entire BCH commutator tower. The “closure” discussed in Sec. manifests here as the fact that χ is a finite scalar number, preventing the proliferation of independent operator directions. All nonlinearities in the coupling strength are packaged into the transcendental dependence of χ and $\mathbf{r}$ on the channel coefficients $\Delta_{x,y,z}$.

Appendix C: Exact Qubit Solution in the Ladder Basis

We provide the manual derivation of the exact qubit mean-force state using the ladder basis $\sigma_\pm$. Starting from the interaction-picture operator product

$$\begin{aligned} \tilde{f}(\tau) \tilde{f}(\tau') &= \left[c^2 + s^2 \cosh(\omega_q(\tau - \tau')) \right] \mathbb{I} \\ &+ s^2 \sinh(\omega_q(\tau - \tau')) \sigma_z \\ &+ cf_- \left(e^{\omega_q \tau'} - e^{\omega_q \tau} \right) \sigma_+ \\ &+ cf_+ \left(e^{-\omega_q \tau} - e^{-\omega_q \tau'} \right) \sigma_-, \end{aligned} \quad (\text{C1})$$

the influence exponent $\Delta = \int_0^\beta d\tau \int_0^\tau d\tau' K(\tau - \tau') \tilde{f}(\tau) \tilde{f}(\tau')$ evaluates to

$$\Delta = \Delta_0\mathbb{I} + \Delta_z\sigma_z + \Delta_+\sigma_+ + \Delta_-\sigma_-, \quad (\text{C2})$$

with coefficients

$$\Delta_z = s^2 \Sigma_z, \quad (\text{C3})$$

$$\Delta_\pm = -2cf_\mp \Sigma_\pm, \quad (\text{C4})$$

where $f_{\pm} = se^{\pm i\phi_f}$ and the primitive integrals are

$$\Sigma_z = \frac{1}{2} [R(\omega_q) - R(-\omega_q)], \quad (\text{C5})$$

$$\Sigma_{\pm} = \frac{1}{\omega_q} [(1 + e^{\pm\beta\omega_q})\mathcal{K}(0) - 2\mathcal{K}(\pm\omega_q)]. \quad (\text{C6})$$

The unnormalised state is $\bar{\rho}_Q = e^{-(\beta\omega_q/2)\sigma_z} e^{\Delta}$. To find the normalized state and its Bloch vector, we perform a symmetric decomposition

$$\bar{\rho}_Q = \Pi^{1/2} e^S \Pi^{1/2}, \quad S = \Pi^{1/2} \Delta \Pi^{-1/2} \quad (\text{C7})$$

where $\Pi = e^{-\beta\omega_q\sigma_z/2}$. Using $\Pi^{1/2}\sigma_{\pm}\Pi^{-1/2} = e^{\mp\beta\omega_q/2}\sigma_{\pm}$, the symmetric influence operator S becomes

$$S = \Delta_0 \mathbb{I} + \Delta_z \sigma_z + (\Delta_+ e^{-\beta\omega_q/2} \sigma_+ + \Delta_- e^{\beta\omega_q/2} \sigma_-). \quad (\text{C8})$$

Substituting the expressions for $\Delta_{\pm}$ and using the KMS relation

$$\Sigma_- = e^{-\beta\omega_q} \Sigma_+, \quad (\text{C9})$$

we define the real symmetrised transverse response

$$\Sigma_{\perp} \equiv 2e^{-\beta\omega_q/2} \Sigma_+ = 2e^{\beta\omega_q/2} \Sigma_-. \quad (\text{C10})$$

The transverse coefficients then simplify to

$$\Delta_+ e^{-\beta\omega_q/2} = -cse^{-i\phi_f} (2e^{-\beta\omega_q/2} \Sigma_+) = -cse^{-i\phi_f} \Sigma_{\perp}, \quad (\text{C11})$$

with $\Delta_{\perp} \equiv -cs\Sigma_{\perp}$. Recombining the ladder operators into Cartesian form, we obtain

$$S = \Delta_0 \mathbb{I} + \Delta_z \sigma_z + \Delta_{\perp} (\sigma_x \cos \phi_f + \sigma_y \sin \phi_f), \quad (\text{C12})$$

which is the generator of a rotation around an axis in the coupling plane. The influence magnitude $\chi = \sqrt{\Delta_z^2 + \Delta_{\perp}^2}$ determines the exponential $e^S = \cosh \chi (\mathbb{I} + \gamma M)$ with $\gamma = \tanh \chi / \chi$ and $M = S - \Delta_0 \mathbb{I}$. The final state $\rho_Q = Z_Q^{-1} \Pi^{1/2} e^S \Pi^{1/2}$ follows by direct matrix multiplication, yielding Eq. (82) in the main text.

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